

Project Description

October 2025

Overview

This document specifies a simplified model of strategic bidding in a liberalized electricity market and proposes a set of exercises that guide you from the bilevel formulation to implementable mixed-integer linear programs (MILPs) and computational experiments.

Deadline: 05/12/2025

Team up in groups of 2-3 people

Hand in: report in pdf and code, include a requirements.txt so all the packages you use are listed.

Problem Setting

Let $\mathcal{J}$ denote the set of strategic players and $\mathcal{I}$ the set of generators in the system. Player $j \in \mathcal{J}$ owns the subset of generators $\Omega_j^{\mathcal{I}} \subseteq \mathcal{I}$. Throughout the exercises you may assume $|\Omega_j^{\mathcal{I}}| = 1$ for clarity, i.e., each player owns one generator, although the notation supports multiple units per player.

Each generator $i \in \mathcal{I}$ can dispatch active power $P_i^{\mathcal{G}}$ within technical bounds

$$P_i^{\mathcal{Gmin}} \leq P_i^{\mathcal{G}} \leq P_i^{\mathcal{Gmax}} \quad \forall i \in \mathcal{I}, \quad (1)$$

to collectively serve an inelastic system demand $D > 0$. The unit marginal production cost is $c_i^{\mathcal{G}}$.

Each generator submits a single price bid $\alpha_i^{\mathcal{G}}$ that must lie within market bounds

$$\alpha^{\min} \leq \alpha_i^{\mathcal{G}} \leq \alpha^{\max} \quad \forall i \in \mathcal{I}. \quad (2)$$

Market clearing (lower level). Given bids $\alpha^{\mathcal{G}}$, the system operator clears the market by solving the economic dispatch:

$$\min_{X_{LL}} \sum_{i \in \mathcal{I}} \alpha_i^{\mathcal{G}} P_i^{\mathcal{G}} \quad \text{s.t.} \quad \sum_{i \in \mathcal{I}} P_i^{\mathcal{G}} = D, \quad P_i^{\mathcal{Gmin}} \leq P_i^{\mathcal{G}} \leq P_i^{\mathcal{Gmax}} \quad \forall i \in \mathcal{I}, \quad (3)$$

with dual variables $\lambda_{n,t}$ (power balance) and $\mu_i^{\text{Gmin}}, \mu_i^{\text{Gmax}}$ (capacity bounds).

Strategic behavior (upper level). Each player $j \in \mathcal{J}$ chooses its bid(s) $\{\alpha_i^{\text{G}}\}_{i \in \Omega_j^{\text{I}}}$ to maximize profit, i.e., revenue at the cleared price minus production cost. For a single-generator player j this yields the canonical bilevel problem:

$$\min_{X_{\text{UL}}, \Xi_{\text{UL}}} \left\{ \sum_{i \in \Omega_j^{\text{I}}} (-\lambda P_i^{\text{G}} + c_i^{\text{G}} P_i^{\text{G}}) \right\} \quad (4a)$$

subject to:

$$\alpha^{\text{min}} \leq \alpha_i^{\text{G}} \leq \alpha^{\text{max}} \quad \forall i \in \Omega_j^{\text{I}}, \quad (4b)$$

$$\min_{X_{\text{LL}}} \left\{ \sum_{i \in \mathcal{I}} \alpha_i^{\text{G}} P_i^{\text{G}} \right\} \quad (4c)$$

subject to:

$$D - \sum_i P_i^{\text{G}} = 0 : \lambda, \quad (4d)$$

$$P_i^{\text{Gmin}} \leq P_i^{\text{G}} \leq P_i^{\text{Gmax}} : \mu_i^{\text{Gmin}}, \mu_i^{\text{Gmax}}, \quad \forall i \in \mathcal{I}. \quad (4e)$$

Where:

$$\begin{aligned} X_{\text{UL}} &= \left\{ \alpha_{i \in \Omega_j^{\text{I}}}^{\text{G}}, P_{i \in \mathcal{I}}^{\text{G}} \right\} \\ \Xi_{\text{UL}} &= \left\{ \lambda_{n,t}, \mu_i^{\text{Gmin}}, \mu_i^{\text{Gmax}} \right\} \\ X_{\text{LL}} &= \left\{ P_{i \in \mathcal{I}}^{\text{G}} \right\} \end{aligned}$$

KKT reformulation. The lower-level problem in (4) is a linear program. It can be replaced by its Karush–Kuhn–Tucker (KKT) optimality conditions—stationarity, primal feasibility, dual feasibility, and complementary slackness—yielding a single-level mathematical program with complementarity constraints (MPCC). Complementarities can be linearized via either a big- M approach with binary variables or SOS1 constraints; the resulting model is a MILP once all bilinear terms are handled.

1 Exercise 1 (15 pt): Single-level MILP reformulation

- Derive the KKT conditions of the lower-level dispatch and replace the lower-level problem in (4) with these conditions to obtain a single-level MPCC.
- Linearize all complementary slackness relations using either (i) big- M with well-justified bounds, or (ii) SOS1 constraints. Explain your choice and the implications for tightness.

- c) If your upper-level objective introduces bilinear terms, introduce auxiliary variables and apply linearizations, so that the final model is a MILP.
- d) Organize your write-up in subsections: model derivation, complementarity handling, linearization details. Include a short *Nomenclature* listing all sets, parameters, variables, and dual variables used.

2 Exercise 2 (15 pt): Single-player experiments

Consider $\mathcal{I} = \{1, 2, 3\}$ with the following guidelines:

- Choose capacities so that $P_i^{\text{Gmin}} = 0$ for all units and P_i^{Gmax} values are small enough that all units are dispatched at the solution, but avoid the knife-edge case $D = \sum_{i \in \mathcal{I}} P_i^{\text{G}}$ at bounds.
- Take $\alpha^{\text{min}} < 0$ and $\alpha^{\text{max}} > 0$ broad enough not to be binding in typical cases.
- Let demand D be high enough so all units receive nonzero dispatch in merit order.

Solve (4) for a *single playing* player j in each of the scenarios below; other players' bids are fixed as indicated.

1. **Truthful competitors:** For all $i \notin \Omega_j^{\text{I}}$, set $\alpha_i^{\text{G}} = c_i^{\text{G}}$.
2. **Overbidding competitors:** For all $i \notin \Omega_j^{\text{I}}$, set $\alpha_i^{\text{G}} > c_i^{\text{G}}$ (justify your choice).

For each scenario, report:

- a) A merit-order plot showing bids (prices), dispatch levels, and the demand intercept.
- b) The playing player's profit.

3 Exercise 3 (15 pt): Best-Response (Diagonalization) among all players

Using the set-up of Exercise 2, compute a Nash equilibrium via iterative best responses ("Diagonalization Algorithm"). Initialize $\alpha_i^{\text{G}} = c_i^{\text{G}}$ for all i at iteration $k = 0$. At iteration $k \geq 1$, update each player's bids by solving their bilevel problem while holding others' bids fixed at iteration $k - 1$.

Stopping criteria. Declare convergence when (i) no player can improve their own profit by more than 1% with a unilateral deviation, *and* (ii) bids stabilize within a user-chosen tolerance. Impose a maximum iteration cap suitable for your machine; if the cap is reached, stop and clearly label the outcome as "not converged".

Deliverables.

- Identify one configuration of $(c_i^G, P_i^{\text{Gmax}}, D, \alpha^{\min}, \alpha^{\max})$ for which the algorithm converges.
- Plot, for each player, the sequence $k \mapsto \alpha_i^G(k)$ together with the truthful cost c_i^G and the final (converged) bid profile.
- Plot the dispatch of each generator at (i) the converged equilibrium and (ii) the centralized optimum obtained by clearing with true costs (“central dispatch”). Use bar charts.
- Compute the inefficiency of the equilibrium respect the central dispatch, report both numerator and denominator explicitly.

4 Exercise 4 (15 pt): Varying fuel costs

Using the convergent set-up identified in Exercise 3:

- For a system with exactly three players, compute the worst equilibrium inefficiency over a bounded *discrete* grid of c_i^G values per generator. State the grid and bounds.
- Repeat the analysis for systems with $|\mathcal{I}| \in \{3, 4, \dots, 10\}$ players/generators. Clearly define how capacities and demand scale with $|\mathcal{I}|$.

Visualize at least:

- a) Equilibrium inefficiency as a function of the number of players $|\mathcal{I}|$.
- b) The share (in %) of instances that converged before hitting the iteration cap.

5 Exercise 5 (40 pt): Open-ended investigation

Choose **one** topic and perform a thorough, well-argued analysis (model choices, parameterization, diagnostics, and discussion):

- Impact of market price bounds $(\alpha^{\min}, \alpha^{\max})$ on the inefficiency of the equilibrium that can be obtained via the Best Response algorithm.
- Sensitivity of the inefficiency of the equilibrium respect to demand level D (e.g., low, medium, peak).
- Effect of asymmetric capacities or costs across players on convergence and inefficiency of the equilibrium.
- Comparison of big- M versus SOS1 linearizations in terms of tightness and runtime.

- Policy-equilibrium concept. How does it differ from the bids-equilibrium, and how policies perform.
- Any other topic you find relevant. Propose and motivate it. Needs to be approved before it can be developed.

Nomenclature (suggested)

Sets: $\mathcal{J}$ (players), $\mathcal{I}$ (generators), $\Omega_j^I \subseteq \mathcal{I}$ (units owned by j).

Parameters: D (demand), c_i^G (marginal cost), $P_i^{G_{\min}}, P_i^{G_{\max}}$ (capacity bounds), $\alpha^{\min}, \alpha^{\max}$ (market price bounds).

Decision variables: P_i^G (dispatch), α_i^G (bid).

Dual variables: $\lambda_{n,t}$ (balance), $\mu_i^{G_{\min}}, \mu_i^{G_{\max}}$ (capacity bounds).

Implementation tips. Use consistent units; document all bounds used in big- M formulations; seed random experiments for reproducibility; and report solver, version.